MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 02** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. Which of the following equations has the sum of its roots equal to 3? [1]

(A) $2x^2 - 3x + 6 = 0$ **(B)** $-x^2 + 3x - 3 = 0$

(C) $\sqrt{2}x^2 - \frac{3}{\sqrt{2}}x + 1 = 0$ **(D)** $3x^2 - 3x + 3 = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** $-x^2 + 3x - 3 = 0$

We know that for a quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$, the sum of roots is given by:

Formula Used: Sum of roots $(\alpha + \beta) = -\frac{b}{a}$

Calculating $-\frac{b}{a}$ for each option:

(A) $-\frac{-3}{2} = \frac{3}{2} \neq 3$

(B) $-\frac{3}{-1} = 3$ (Matches the required sum!)

(C) $-\frac{-3/\sqrt{2}}{\sqrt{2}} = \frac{3}{2} \neq 3$

(D) $-\frac{-3}{3} = 1 \neq 3$

Therefore, option **(B)** is correct. **[1 mark]**

Q2. Values of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots are: [1]

(A) 0 only **(B)** 4

(C) 8 only **(D)** 0, 8

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) 0, 8

Comparing $2x^2 - kx + k = 0$ with $ax^2 + bx + c = 0$, we get $a = 2$, $b = -k$, and $c = k$.

Note that $a = 2 \neq 0$ for all values of k.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

Setting $D = 0$:

$$(-k)^2 - 4(2)(k) = 0 \implies k^2 - 8k = 0 \implies k(k-8) = 0 \implies \boxed{k = 0 \text{ or } k = 8}$$

Both $k = 0$ (equation $2x^2 = 0$) and $k = 8$ ($2x^2 - 8x + 8 = 0$) give equal roots. Hence option **(D)** is correct. **[1 mark]**

Q3. Which of the following equations has two distinct real roots? [1]

(A) $2x^2 - 3\sqrt{2}x + \frac{9}{4} = 0$ **(B)** $x^2 + x - 5 = 0$

(C) $x^2 + 3x + 2\sqrt{2} = 0$ **(D)** $5x^2 - 3x + 1 = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $x^2 + x - 5 = 0$

We know that a quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots if and only if $D > 0$.

Formula Used: Two distinct real roots $\iff D = b^2 - 4ac > 0$

Checking D for each option:

(A) $D = (-3\sqrt{2})^2 - 4(2)\left(\frac{9}{4}\right) = 18 - 18 = 0$ (equal real roots).

(B) $D = (1)^2 - 4(1)(-5) = 1 + 20 = 21 > 0$ (two distinct real roots).

(C) $D = (3)^2 - 4(1)(2\sqrt{2}) = 9 - 8\sqrt{2} \approx 9 - 11.31 < 0$ (no real roots).

(D) $D = (-3)^2 - 4(5)(1) = 9 - 20 = -11 < 0$ (no real roots).

Hence, option **(B)** is correct. **[1 mark]**

Q4. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has: [1]

(A) two distinct real roots **(B)** two equal real roots

(C) no real roots **(D)** more than 2 real roots

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) no real roots

Comparing $2x^2 - \sqrt{5}x + 1 = 0$ with $ax^2 + bx + c = 0$, we have $a = 2$, $b = -\sqrt{5}$, and $c = 1$.

Formula Used: $D = b^2 - 4ac < 0 \implies$ No real roots

Computing the discriminant D:

$$D = (-\sqrt{5})^2 - 4(2)(1) = 5 - 8 = -3 < 0$$

Since $D < 0$, the equation has no real roots. Thus option **(C)** is correct. **[1 mark]**

Q5. Assertion-Reason: [1]

Assertion (A): The equation $x^2 + 2x + 5 = 0$ has no real roots.

Reason (R): If the discriminant $D = b^2 - 4ac$ of a quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ is negative $(D < 0)$, then the equation has no real roots.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

For $x^2 + 2x + 5 = 0$, $a = 1$, $b = 2$, and $c = 5$.

Formula Used: $D = b^2 - 4ac < 0 \implies$ No real roots

Computing D:

$$D = (2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$$

Because $D < 0$, the square root $\sqrt{D}$ in the quadratic formula is not a real number, so $x^2 + 2x + 5 = 0$ has no real roots. Hence Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Section B: Short Answer Questions

MATHS MANTHAN | **2 + 3 + 3 + 3 = 11 Marks**

Q6. Find the roots of the quadratic equation $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorisation. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given quadratic equation: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$.

Here $a = \sqrt{2}$, $b = 7$, and $c = 5\sqrt{2}$.

Formula Used: $ac = (\sqrt{2})(5\sqrt{2}) = 5 \times 2 = 10, \quad b = 7$

We split $7 = 2 + 5$ because $2 \times 5 = 10$ and $2 + 5 = 7$. **[$\frac{1}{2}$ mark]**

Writing $2x = \sqrt{2} \cdot \sqrt{2}\,x$:

$$\sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = 0$$
$$\sqrt{2}x(x + \sqrt{2}) + 5(x + \sqrt{2}) = 0 \quad \textbf{[1 mark]}$$
$$(x + \sqrt{2})(\sqrt{2}x + 5) = 0$$

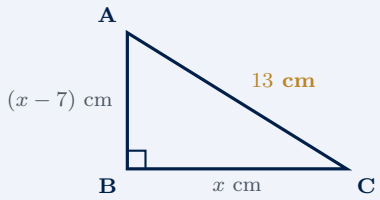
Equating each factor to zero:

$$x + \sqrt{2} = 0 \implies x = -\sqrt{2} \qquad \text{or} \qquad \sqrt{2}x + 5 = 0 \implies x = -\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}$$

Hence, the roots are $\boxed{x = -\sqrt{2} \text{ and } x = -\frac{5\sqrt{2}}{2}}$. **[$\frac{1}{2}$ mark]**

Q7. The altitude of a right-angled triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Let the base of the right-angled triangle ABC (right-angled at B) be $BC = x$ cm ($x > 7$). Then its altitude is $AB = (x-7)$ cm and hypotenuse is $AC = 13$ cm, as shown alongside.

Formula Used: Pythagoras' Theorem: $(\text{Base})^2 + (\text{Altitude})^2 = (\text{Hypotenuse})^2$

Substituting the lengths of the sides of $\triangle ABC$ ($\angle B = 90°$):

$$x^2 + (x-7)^2 = 13^2$$
$$x^2 + (x^2 - 14x + 49) = 169 \quad \textbf{[1 mark]}$$
$$2x^2 - 14x - 120 = 0$$
$$x^2 - 7x - 60 = 0 \quad [\tfrac{1}{2} \textbf{ mark}]$$

Factorising using $-12 \times 5 = -60$ and $-12 + 5 = -7$:

$$x^2 - 12x + 5x - 60 = 0 \implies x(x-12) + 5(x-12) = 0$$
$$\implies (x-12)(x+5) = 0 \implies x = 12 \text{ or } x = -5 \quad \textbf{[1 mark]}$$

Since length cannot be negative ($x > 7 > 0$), we reject $x = -5$.
Therefore, Base $= x = 12$ cm and Altitude $= x - 7 = 12 - 7 = 5$ cm.
Hence, the other two sides are $\boxed{\text{Base} = 12 \text{ cm and Altitude} = 5 \text{ cm}}$. $[\tfrac{1}{2} \textbf{ mark}]$

Q8. Find the roots of the quadratic equation $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$ by factorisation. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given quadratic equation: $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$.
Here $a = \sqrt{3}$, $b = 10$, and $c = -8\sqrt{3}$.

Formula Used: $ac = (\sqrt{3})(-8\sqrt{3}) = -24, \quad b = 10$

Two numbers whose sum is 10 and product is -24 are 12 and -2 ($12 - 2 = 10$, $12 \times (-2) = -24$). **[1 mark]**
Writing $12 = 4\sqrt{3} \cdot \sqrt{3}$ and splitting the middle term:

$$\sqrt{3}x^2 + 12x - 2x - 8\sqrt{3} = 0$$
$$\sqrt{3}x(x + 4\sqrt{3}) - 2(x + 4\sqrt{3}) = 0 \quad \textbf{[1 mark]}$$
$$(x + 4\sqrt{3})(\sqrt{3}x - 2) = 0$$

Equating each factor to zero:

$$x + 4\sqrt{3} = 0 \implies x = -4\sqrt{3} \qquad \text{or} \qquad \sqrt{3}x - 2 = 0 \implies x = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

Hence, the roots are $\boxed{x = -4\sqrt{3} \text{ and } x = \frac{2\sqrt{3}}{3}}$. **[1 mark]**

Q9. The difference of squares of two numbers is 180. The square of the smaller number is 8 times the larger number. Find the two numbers. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let the larger number be x and the smaller number be y.

Formula Used: $x^2 - y^2 = 180$ and $y^2 = 8x$

Substituting $y^2 = 8x$ into the first equation:

$$x^2 - 8x = 180 \implies x^2 - 8x - 180 = 0 \quad \textbf{[1 mark]}$$

Factorising by splitting the middle term ($-18 \times 10 = -180$ and $-18 + 10 = -8$):

$$x^2 - 18x + 10x - 180 = 0 \implies x(x-18) + 10(x-18) = 0$$
$$\implies (x-18)(x+10) = 0 \implies x = 18 \text{ or } x = -10 \quad \textbf{[1 mark]}$$

Case 1: If $x = -10$, then $y^2 = 8(-10) = -80 < 0$. Since the square of a real number cannot be negative, $x = -10$ is rejected.
Case 2: If $x = 18$, then $y^2 = 8(18) = 144 \implies y = \pm\sqrt{144} = \pm 12$.
Notice that both $y = 12$ and $y = -12$ are smaller than $x = 18$ and satisfy $18^2 - (\pm 12)^2 = 324 - 144 = 180$.
Hence, the two pairs of numbers are $\boxed{(18,\ 12) \text{ and } (18,\ -12)}$. **[1 mark]**

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Q10. Case Study: Eco-Tourism River Safari & Upstream–Downstream Dynamics

An eco-tourism motorboat whose speed in still water is 20 km/h takes a safari group 48 km upstream and returns 48 km downstream to the same jetty. The total time for the round trip is 5 hours. Let the speed of the river stream be x km/h ($0 < x < 20$). [4]

(i) Write the speeds of the boat going downstream and going upstream in terms of x. **[1]**

(ii) Form and simplify the quadratic equation in x representing the total journey time of 5 hours. **[1]**

(iii) Find the speed of the river stream. **OR** Find the time taken separately for the upstream and downstream journeys. **[2]**

► **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

We know the upstream and downstream speed rules for a boat moving in a stream of speed x:

Formula Used: Downstream speed $= u + x$, Upstream speed $= u - x$, Time $= \dfrac{\text{Distance}}{\text{Speed}}$

(i) Effective Speeds: Here speed in still water $u = 20$ km/h and stream speed $= x$ km/h ($0 < x < 20$).

$\boxed{\text{Downstream speed} = (20 + x) \text{ km/h}}$ and $\boxed{\text{Upstream speed} = (20 - x) \text{ km/h}}$ **[1 mark]**

(ii) Quadratic Equation: Total time for 48 km upstream and 48 km downstream is 5 hours:

$$\frac{48}{20 - x} + \frac{48}{20 + x} = 5$$

$$48\left[\frac{(20 + x) + (20 - x)}{(20 - x)(20 + x)}\right] = 5$$

$$\frac{48 \times 40}{400 - x^2} = 5 \implies 400 - x^2 = \frac{1920}{5} = 384 \implies \boxed{x^2 - 16 = 0}$$ **[1 mark]**

(iii) First Part (Speed of Stream): Solving $x^2 - 16 = 0$:

$$(x - 4)(x + 4) = 0 \implies x = 4 \text{ or } x = -4$$

Since speed of the stream is positive ($x > 0$), we reject $x = -4$.

Hence, the speed of the river stream is $\boxed{4 \text{ km/h}}$. **[2 marks]**

OR (Alternative Part — Upstream & Downstream Times): Using $x = 4$ km/h from part (iii):

Upstream speed $= 20 - 4 = 16$ km/h, Downstream speed $= 20 + 4 = 24$ km/h

Therefore:

$$t_{\text{upstream}} = \frac{48}{16} = 3 \text{ hours}, \qquad t_{\text{downstream}} = \frac{48}{24} = 2 \text{ hours}$$

Verification: $3 + 2 = 5$ hours. Hence, $\boxed{\text{Upstream time} = 3 \text{ hours}}$ and $\boxed{\text{Downstream time} = 2 \text{ hours}}$. **[2 marks]**

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